THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(d)** 140 degrees	Initial i = 55; new i = 70; angle = 140.
6	**(d)** 15 m	3/4 = h/20.
7	**(b)** snake	Energy decreases at each step.
8	**(c)** 0.53	3400/6400.
9	**(d)** 90 degrees to the surface (back along normal)	Incidence along the normal.
10	**(b)** about 400 times farther	Equal angular sizes.
11	**(a)** bees and flowers	Both benefit.
12	**(b)** flooding by rising sea levels	Higher sea level.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
15	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Initial i = 60 degrees (1); New i = 70 or 50 degrees (1).	1+1
17	365.25 is slightly more than the actual 365.2422 (1); 1900 is not a leap year but 2000 is (1).	1+1
18	Grass is the producer; deer the primary consumer; tiger the secondary consumer (top carnivore) (1); The tiger level has the least energy (1).	1+1
19	0.15 x 1.08 x 10^12 (1); = 1.6 x 10^11 cubic km (1).	1+1
20	Size depends on reflection geometry, not distance (1); Image is as big as the object (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	n = 360/theta - 1 = 7 (1); 360/theta = 8 (1); theta = 45 degrees (1).	1+1+1
22	At noon: almost no shadow (1); At 3 pm: a shadow pointing east (1); Length is larger than at noon (1).	1+1+1
23	Energy decreases at each trophic level (1); A small amount of energy reaches the top level (1); So only a few top carnivores can be supported (1).	1+1+1
24	Dense iron sank to the centre (1); Lighter material floated up (1); Seismic S-waves do not pass through the outer core (1).	1+1+1
	OR: Close: virtual, erect and enlarged image (1); Far: real and inverted image (1); So the face appears inverted at a distance (1).	1+1+1
25	Earth rotates 360 degrees in 24 hours (1); 15 degrees per hour (1); Local noon differs with longitude (1).	1+1+1
	OR: Fertiliser run-off feeds algae (1); Algae die and decay using oxygen (1); Fish die and water stinks (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	f = R/2 (1); = 20/2 = 10 cm (1); Sun's rays meet at the principal focus, 10 cm from the pole (1); Bulb is placed at the focus (1); Reflected rays form a parallel beam (1).	1+1+1+1+1
	OR: Measure the shadow of a vertical stick at local noon each week (1); Longest in December (1); Shortest in June (1); Medium at equinoxes (1); Graph shows a smooth annual cycle (1).	1+1+1+1+1
27	Herbivores increase (1); Overgrazing of grass (1); Soil erosion (1); Water and shelter loss (1); Ecosystem degrades; other species decline (1).	1+1+1+1+1
	OR: Inner core = 6400 - (35 + 2900 + 2200) (1); = 1265 km (1); Crust 0.5 per cent, mantle 45 per cent (1); Outer core 34 per cent, inner core 20 per cent (1); Mantle is the thickest (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 65 degrees	i2 = 90 - 25.
3	**(d)** decreased	Longer shadow means a lower Sun.
4	**(c)** support all the other levels and hold the most energy	All other organisms depend on producers.
5	**(c)** 42 million cubic km	0.03 x 1.4 billion.
6	**(b)** parallel	Parallel beam.
7	**(a)** about 3 years	11 days x 3 = 33 days.
8	**(d)** parasitism	One benefits, one harmed.
9	**(d)** using public transport and saving electricity	Reduces emissions.
10	**(c)** A is true but R is false.	A is true; R is false because they form virtual diminished images.
11	**(c)** A is true but R is false.	A is true; R is false because it rotates once per revolution.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(b)** 8 m	Boy is 5 m from the mirror; image 5 m behind; man at 3 m; total 8.
15	**(a)** high in the sky	Lunar eclipse occurs on a full moon, which is high in the sky at midnight.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Its lit half faces away from the Earth (1); Dark side faces us (1).	1+1
17	Trees take in CO2 for photosynthesis (1); Carbon is stored in wood (1).	1+1
18	Air is extremely thin (1); Gases gradually escape to space (1).	1+1
19	It bends parallel rays towards each other (1); They meet at the principal focus (1).	1+1
20	Night side faces different directions (1); Different constellations are visible (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Early atmosphere had CO2 and water vapour (1); Photosynthesis added oxygen (1); Ozone layer formed and life spread on land (1).	1+1+1
22	Both use convex lenses (1); Camera uses a film or sensor; eye uses retina (1); Both give real inverted images (1).	1+1+1
23	Circumference about 940 million km (1); Time = 365.25 x 24 x 3600 s = 3.16 x 10^7 s (1); Speed about 30 km/s (1).	1+1+1
24	Producing meat uses more plant energy (1); Energy lost at each step (1); Plant diets are efficient (1).	1+1+1
	OR: Day length would be about 12 hours (1); Shorter days and nights (1); Winds and weather would change (1).	1+1+1
25	Image is 15 cm behind the mirror (1); Distance from object = 30 cm (1); New: object at 11 cm, so distance = 22 cm (1).	1+1+1
	OR: Difference 0.0078 x 100 (1); = 0.78 day (1); Three leap days omitted every 400 years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Cornea and lens focus light (1); Image forms on retina (1); Myopia: concave lens (1); Hypermetropia: convex lens (1); Cataract and other causes (1).	1+1+1+1+1

	OR: Latitude = 30 degrees N (1); Equator: on the horizon (1); North Pole: overhead (1); Height increases as he travels north (1); Pole Star altitude equals latitude (1).	1+1+1+1+1
27	Global warming (1); Ice melts and sea level rises (1); Ocean acidification (1); Extreme weather (1); Individual: save energy and plant trees (1).	1+1+1+1+1
	OR: Atmosphere: warming (1); Oceans: more acidic, coral damage (1); Plants: faster growth up to a point (1); But heat and drought reduce crops (1); Net effect is harmful (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** increase	Fewer predators.
2	**(c)** pressure raises the melting point	High pressure keeps it solid.
3	**(d)** an erect, diminished image that covers a wide field	A convex mirror gives a wide field of view.
4	**(c)** one year	33 x 11 = 363 days.
5	**(b)** algae and fungi	Mutual benefit.
6	**(a)** the water table falls	Overdraft.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** A is true but R is false.	A is true; R is false: it reflects sunlight.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** A is true but R is false.	A is true; R is false: the Moon has almost no atmosphere.
11	**(a)** 1:50	12:00 - 10:10.
12	**(a)** 15 days	Half month.
13	**(a)** a decrease in dissolved oxygen after the algae die	Eutrophication.
14	**(c)** Venus	Runaway greenhouse.
15	**(d)** smaller and closer to F	Diminished virtual image nears F.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Plants form the base (1); Energy decreases upward (1).	1+1
17	Air is thinner (1); Less heat is held, and air cools with height (1).	1+1
18	Image moves at 3 m/s away from the mirror (1); Relative speed = 6 m/s (1).	1+1
19	Ratio 1.5/2 = h/40 (1); h = 30 m (1).	1+1
20	Birds eat poisoned insects (1); Poison passes along the food chain (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Pollinators die, less crop yield (1); Predators of pests die (1); Birds starve (1).	1+1+1
22	70/6400 x 100 (1); = 1.1 per cent (1); Crust is a very thin skin (1).	1+1+1
23	Concave reflector with bulb at F (1); Reflected rays become parallel (1); Gives a strong narrow beam (1).	1+1+1
24	Difference 7.5 degrees (1); 7.5 / 15 = 0.5 hour (1); 30 minutes behind IST (1).	1+1+1
	OR: Roots hold soil (1); Leaf litter adds humus (1); Animals and decomposers return (1).	1+1+1
25	Lithosphere: soil and minerals (1); Hydrosphere: water (1); Atmosphere: oxygen and CO2 (1).	1+1+1
	OR: Convex: virtual, erect (1); Always smaller (1); Image lies behind the mirror (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	First mirror: i = r = 35 degrees; the ray makes 55 degrees with the mirror surface (1); In the triangle with the right angle, the ray makes 35 degrees with the second mirror surface (1); Angle of incidence at the second mirror = 55 degrees (1); The emergent ray turns through 180 degrees in all (1); Hence it is parallel to the incident ray but opposite in direction (1).	1+1+1+1+1

	OR: Divisible by 4 is leap (1); Century years need divisibility by 400 (1); 1800, 1900, 2100 not leap (1); 2000 and 2024 leap (1); Average year length 365.2425 days (1).	1+1+1+1+1
27	Chains: grass, grasshopper, frog, snake, hawk (1); Grass, rabbit, hawk; grass, mouse, snake, hawk (1); Hawk: third, fourth and fifth levels (1); Without frogs: grasshoppers rise, snakes switch to mice (1); Network adjusts but balance is changed (1).	1+1+1+1+1
	OR: Venus: strong greenhouse and closer to Sun (1); Earth: moderate greenhouse (1); Mars: thin atmosphere, farther from Sun (1); Atmosphere composition matters (1); Balance is delicate for habitability (1).	1+1+1+1+1

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